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StyleDomain: Efficient and Lightweight Parameterizations of StyleGAN for One-shot and Few-shot Domain Adaptation

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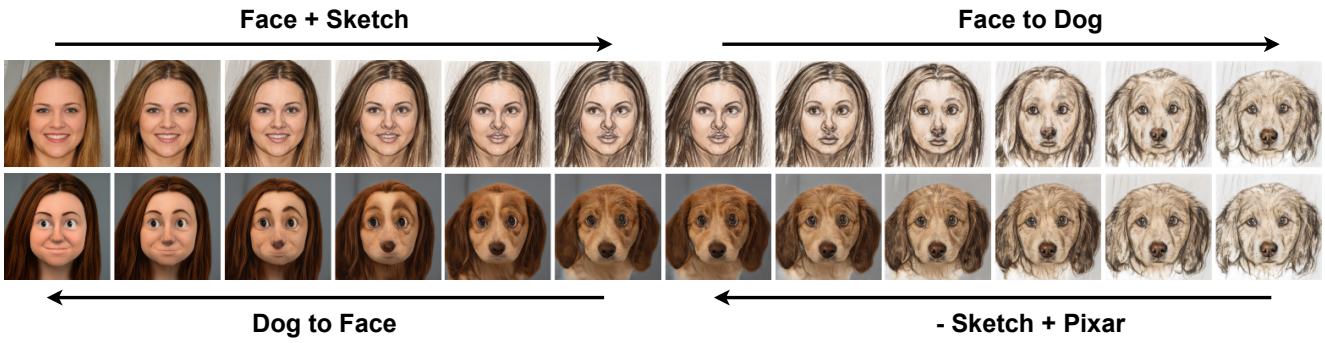


Figure 1: Smooth cross-domain image morphing. Morphing in the space of generator weights could be successfully combined with morphing using StyleDomain directions, i.e. directions in the StyleSpace that can adapt the generator to new domains (see Section 4).

Abstract

Domain adaptation of GANs is a problem of fine-tuning GAN models pretrained on a large dataset (e.g. StyleGAN) to a specific domain with few samples (e.g. painting faces, sketches, etc.). While there are many methods that tackle this problem in different ways, there are still many important questions that remain unanswered. In this paper, we provide a systematic and in-depth analysis of the domain adaptation problem of GANs, focusing on the StyleGAN model. We perform a detailed exploration of the most important parts of StyleGAN that are responsible for adapting the generator to a new domain depending on the similarity between the source and target domains. As a result of this study, we propose new efficient and lightweight parameterizations of StyleGAN for domain adaptation. Particularly, we show that there exist directions in StyleSpace (StyleDomain directions) that are sufficient for adapting to similar domains. For dissimilar domains, we propose Affine+ and AffineLight+ parameterizations that allows us to outperform existing baselines in few-shot adaptation while having significantly less training parameters. Finally, we examine StyleDomain directions and discover their many surprising properties that we apply for domain mixing and cross-domain image morphing. Source code can be found at https://github.com/AIRI-Institute/StyleDomain.

1. Introduction

Recent years GANs [12, 20, 21, 19, 5] have shown impressive results in image synthesis and offered many ways to control the generated data. In particular, the state-of-the-art StyleGAN models [20, 21, 19] have many practical applications such as image enhancement [46, 24, 6, 41], image editing [17, 35, 14, 37, 1, 43, 29], image-to-image translation [31, 16, 36, 11] thanks to their high-quality image generation and their latent representation that has rich semantics and disentangled controls for localized meaningful image manipulations. However, it comes at a price, as the training of StyleGAN requires a large, high-quality dataset that significantly limits its applicability because many real-world domains are represented by few images. The standard approach to deal with this problem is transfer learning, i.e. fine-tuning the model pretrained on the source domain A to the target domain B.

There are many domain adaptation methods for StyleGAN [18, 39, 50, 52, 23, 45, 28, 31, 4, 11, 55, 44] that

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tackle this problem in different ways depending on the number of available images (e.g., one-shot/few-shot) from the target domain B and the similarity between the source A and target B domains (e.g., faces $\rightarrow$ sketches, artistic portraits, or faces $\rightarrow$ dogs, cats). Most of these works implicitly assume that StyleGAN can be adapted to a new domain only if we fine-tune almost all its weights, even for similar domains. However, this common wisdom is poorly investigated and verified and there is a lack of analysis of which parts of StyleGAN are important depending on different data regimes and the similarity between domains.

In this work, we aim to provide a systematic and comprehensive analysis of this question. Our investigation of the properties of the aligned StyleGAN models consists of several parts. First, in Section 3, we identify what parts of the StyleGAN are sufficient for its adaptation depending on the similarity between the source A and target B domains. We discover that fine-tuning the whole synthesis convolutional network is not always necessary. In the case of similar A and B domains, the affine layers are sufficient for the adaptation. For more distant domains, we should optimize more parameters, however not the whole network. It suggests investigating new more efficient and lightweight parameterizations of StyleGAN to utilize them for domain adaptation.

In the second part of our analysis, we propose two new parameterizations of StyleGAN. For similar domains, we consider the latent space that is formed by the output of affine layers, i.e. StyleSpace [43]. We show that we can directly optimize directions in this space that can adapt to similar target domains with the same quality as fine-tuning all weights of StyleGAN (we call such directions as *StyleDomain* directions). Further, we explore that we can zero out 80% of StyleDomain direction coordinates without a quality degradation that gives even more lightweight parameterization (*StyleSpaceSparse*). For more distant domains, we propose a new parameterization *Affine+* that consists of affine layers and only one convolutional block from the synthesis network. It reduces the number of trainable parameters by 6 times and achieves the same quality. Then, we further improve Affine+ parameterization by utilizing low-rank decomposition for weights of affine layers and obtain *AffineLight+* parameterization. It allows us to optimize by two orders less parameters compared to training the whole StyleGAN. These parameterizations show the state-of-the-art performance for few-shot adaptation for dissimilar domains outperforming more complicated and expensive baselines.

Additionally, in Section 4, we inspect StyleDomain directions and discover their surprising properties. The first one is mixability, i.e. we can sum up these directions to obtain a new mixed domain (e.g., see Figure 6 as a mix of the Joker style, Pixar style and the style from the image). The second impressive property is transferability, i.e. the same StyleDomain directions can be applied to StyleGAN models that were fine-tuned to other domains (e.g., see Figure 5 where we apply directions found for faces to dogs, cats and churches). We apply these findings to standard computer vision tasks such as image-to-image translation and cross-domain morphing.

2. Related Work

Latent Spaces of StyleGAN. After recent remarkable success of GANs [12, 20, 21, 19, 5] in image synthesis, many works appeared that explore their latent representation for controllable image manipulation. In particular, the latent space of StyleGAN [20, 21, 19] has attracted considerable attention. It consists of three levels: (i) the first latent space, $\mathcal{Z}$, is raw random noise (typically Gaussian); (ii) the intermediate latent spaces $\mathcal{W}, \mathcal{W}+$ [1] are formed by the output of the mapping network; (iii) the last level is StyleSpace, $\mathcal{S}$, [43] that is spanned by the channel-wise style parameters after affine layers. It has been shown that these latent spaces have rich semantics [20, 21], and especially the StyleSpace that demonstrates the most disentangled and localized semantical directions [43]. In recent years many works have proposed to utilize such appealing properties of the StyleGAN latent spaces for image editing tasks [17, 35, 14, 37, 1, 43, 29]. To apply these methods for real images it is necessary to inverse them into one of the latent space of StyleGAN which is another task that draws significant attention [1, 21, 13, 30, 33, 38, 54, 53]. We should note that all mentioned methods for image manipulation by controlling StyleGAN latent space allow only in-domain editing. In this paper, we show that in StyleSpace there exist such directions that can change the domain of images.

Domain Adaptation of StyleGAN. Recent years the problem of fine-tuning StyleGAN has generated a great deal of interest as it allows training the state-of-the-art generative model for a domain with few samples. There have appeared many works that tackle this problem in different ways depending on how similar the target domain is to the source one. Roughly, these methods can be divided into two groups. The first one deals with the case when the target and source domains are dissimilar (e.g. faces $\rightarrow$ cats, churches, etc.). In contrast, the second group considers the setting of similar domains (e.g. faces $\rightarrow$ stylized faces, painting faces, sketches, etc.). Methods from the first group typically require hundreds or thousands of samples from a new domain to adapt faithfully and they leverage data augmentations [18, 39, 50, 52], or freeze some layers of the discriminator to prevent overfitting [25], or train the discriminator with auxiliary losses to match the data more accurately [23, 45].

In the setting of the second group it is sufficient to have

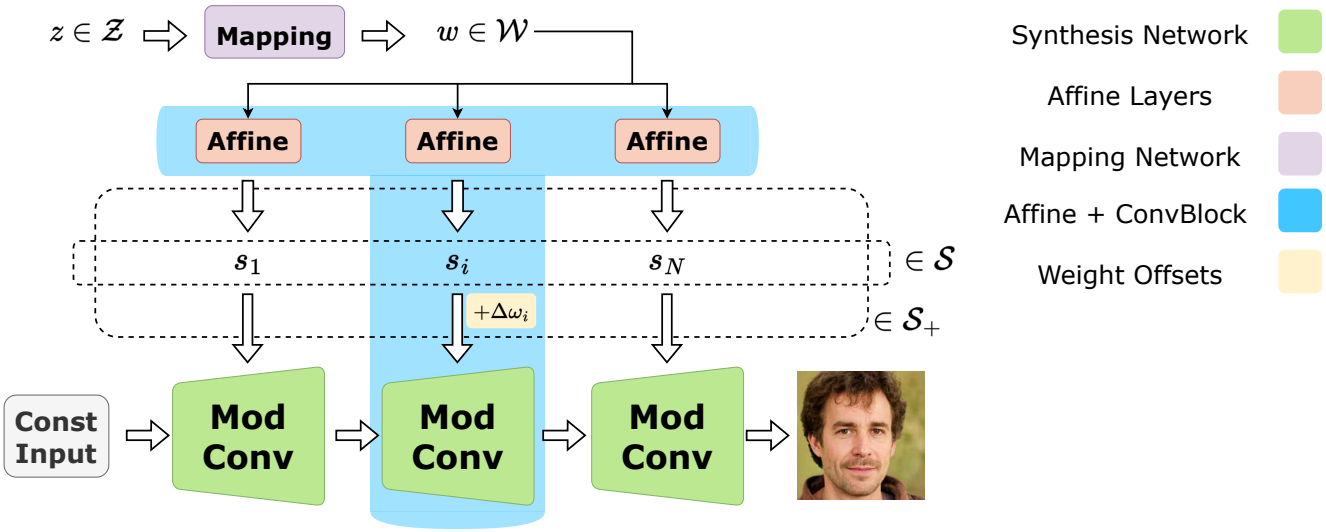


Figure 2: StyleGAN2 architecture. We introduce new latent space $S+$ for the for domain adaptation that combines StyleSpace and weight offsets for one block from the synthesis network.

only several samples (up to dozens) from the target domain for the successful adaptation and such approaches utilize another techniques. In particular, they introduce additional regularization terms [40, 22], preserve pairwise distances between instances in the source domain via cross-domain consistency loss [28], mix weights of the fine-tuned and the base generators [31], use an auxiliary small network to enhance the training [42]. More advanced methods [11, 55, 4] utilize the pretrained CLIP model [32] as the vision-language supervision for the text-based adaptation [11, 4] or the one-shot image-based adaptation [7, 55, 9, 49, 48, 4].

Recent work [51] analyzes the performance of methods from the second group for dissimilar domains in the low data regime. It was shown that those methods have a significant quality degradation in the few-shot regime. However, it can be partially mitigated by constraining model parameters. There are also approaches that introduce more lightweight parameterizations for domain adaptation [27, 34, 7, 4], however they work only for similar domains. In our paper, we propose efficient and highly lightweight parameterizations for both similar and dissimilar domains and show that they can achieve results on par with the state-of-the-art methods that optimize all weights of StyleGAN.

There are few works that analyze the domain adaptation process of StyleGAN thoroughly. The paper [44] is the first attempt to perform such an in-depth study. In particular, they explore which parts of StyleGAN are mostly changed during fine-tuning and the transferability of the latent semantics after domain adaptation. However, this work does not analyze which parts of StyleGAN are sufficient for adapting depending on the similarity between source and target domains. In our paper, we provide a more systematic and comprehensive analysis that completes and improves the results from [44].

3. Importance of Each Part of the StyleGAN

In this section, our goal is to analyze what parts of StyleGAN are important for domain adaptation. Similarly to previous works, we specifically focus on the state-of-the-art GAN architecture, StyleGAN2 [21]. For the source domain, we consider FFHQ [20] as it is the large high-quality dataset that is suitable for training StyleGAN2 from scratch. For the target one, we test a wide range of different domains that we will describe further.

StyleGAN2 structure and its main components. We provide a diagram description of the StyleGAN2 architecture in Figure 2. It consists of three parts:

- *mapping network* f_M that transforms the input noise $z \in \mathcal{Z}$ (typically Gaussian) to the intermediate latent vector $w \in \mathcal{W}$;
- *affine layers* $f_1^A, \ldots, f_N^A$, each of them takes as input the vector w and maps it to corresponding style vector $s_1 = f_1^A(w), \ldots, s_N = f_N^A(w)$. The concatenation of these vectors form the vector from the StyleSpace: $s = (s_1, \ldots, s_N) \in \mathcal{S}$;
- *synthesis network* that is a composition of modulated convolutions. The weights of each convolution are modulated by the input style vector s_i and applied to the input feature maps. The synthesis network also has tRGB convolutional layers that transform feature maps to RGB images and they also are modulated by corresponding style vectors.

Accordingly, the StyleGAN2 G_θ generates from the input noise z the output image $I = G_\theta(s(z))$ where θ are

weights of the synthesis network, $s(z) = f^A(f_M(z))$, $f^A = \{f_1^A, \dots, f_N^A\}$.

We will analyze these three components of StyleGAN2 and their impact on the domain adaptation process. It is common wisdom that the most important part for adaptation is the synthesis network, while the mapping network and affine layers are mostly responsible for the semantic manipulations within the source domain [44]. We aim to verify whether such a conception is correct.

In our experiments, we additionally consider the impact of the combination of affine layers and one convolutional block from the synthesis network on domain adaptation. It is a way to probe the intermediate case between affine layers and the synthesis network.

Method to analyze the impact of each component. The paper [44] has proposed to analyze the impact of each component by resetting its weights in the fine-tuned generator to their pretrained values and assessing the quality of the generated images. In our work, we propose another approach: to directly fine-tune only one of these components to explore which ones are sufficient for domain adaptation.

Let us describe our method in more detail. The optimization process for the problem of domain adaptation is the following

$$\mathcal{L}_D\left(\{G_\theta(s(z_i))\}_{i=1}^K\right) \to \min, \tag{1}$$

where $\mathcal{L}_D$ is domain adaptation loss that depends on the domain D (we discuss it further) and the samples from the generator $\{G_\theta(s(z_i))\}_{i=1}^K$, $z_1, \dots, z_K \in \mathcal{Z}$ are sampled noises.

Typically the generator G_θ is optimized with respect to all components, i.e.

$$\mathcal{L}_D\left(\{G_\theta(s(z_i))\}_{i=1}^K\right) \to \min_{\theta, f^A, f_M}. \tag{2}$$

We propose to investigate settings when we optimize with respect to only one these components. We denote each parameter space as: $\{\theta\}$ – *SyntConv* parameterization, $\{f^A\}$ – *Affine* parameterization, $\{f_M\}$ – *Mapping* parameterization. The case we fine-tune all components of the StyleGAN2 we call *Full* parameterization.

Domain adaptation settings. In our study, we consider two settings: one-shot and few-shot. For each data regime, we use different domains depending on their similarity to the source domain of realistic faces from FFHQ:

- one-shot domains: for this setting, we consider only similar domains. It is the case when the target domain has the same geometry of faces and it preserves the identity of the person. It alters only the style of the image. In this regime, we consider not only one-shot image-based adaptation with the reference stylized face but additionally text-based adaptation with the text description of the target style (e.g., "Photo in the style of anime (pixar, sketch, etc.)"). See examples of such domains in Figure 3.
- few-shot domains: for this regime, we examine more distant domains that have a face-like form but change the face geometry and identity in a stronger manner. As examples, we consider AFHQ dogs faces and cats faces [8].

Depending on the data regime, we use different domain loss function $\mathcal{L}_D$. For one-shot domain adaptation, we apply the optimization loss from StyleGAN-NADA [11] in the case of text-based adaptation and another loss from DiFa [48] for one-shot image-based adaptation. For few-shot domain adaptation, we utilize the fine-tuning procedure from StyleGAN-ADA [18]. For more details about domain adaptation loss functions see Appendix A.1.

To obtain quantitative comparisons in the case of a one-shot setting, we use Quality and Diversity metrics that were proposed in the HyperDomainNet paper [4]. For few-shot adaptation, we compute FID metric [15] using the standard protocol from [18].

Analysis for one-shot domains. For the analysis, we choose different text-based and one-shot image-based domains (see Appendix A.2 for the full list and more details). In experiments, we consider the four parameterizations (Full, SyntConv, Affine, Mapping) we discussed above.

We provide qualitative results in Figure 3 with quantitative ones in Table 1. More results see in Appendix A.2.

We observe that all three parameterizations, Full, SyntConv and Affine, show comparable performance in terms of both visual quality and objective metrics. The fact that the synthesis network is sufficient for similar domains was clear from the previous work [44]. However, our finding that the affine part is also sufficient is a new and surprising result. It means that we can change the domain of generated images without fine-tuning the synthesis network but just passing the modified style vector that comes from the affine part. Also, we observe that the mapping network shows poor vi-

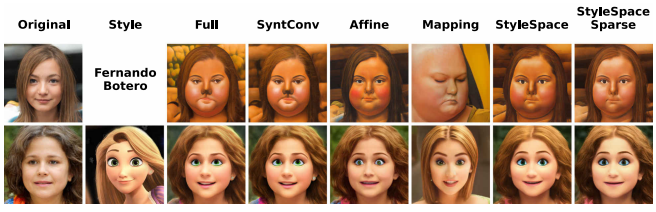


Figure 3: Text-based and image-based adaptation for different parameterizations. Affine, StyleSpace and StyleSpaceSparse parameterizations yield performance comparable with Full one. This style image is called "Disney".

Table 1: Quality and Diversity metrics [4] for text-based and one-shot image-based domain adaptations with different parameterizations. Affine, StyleSpace and StyleSpaceSparse parameterizations achieve results comparable with the Full one.

		Botero		**Sketch**		**Disney (image)**		**Titan Erwin (image)**	
Parameter Space	Size	Quality	Diversity	Quality	Diversity	Quality	Diversity	Quality	Diversity
Full	30.3M	0.312	0.228	0.208	0.296	0.713	0.247	0.760	0.194
SyntConv	23.6M	0.311	0.224	0.191	0.292	0.711	0.259	0.741	0.217
Affine	4.6M	0.298	0.221	0.194	0.296	0.565	0.359	0.650	0.314
Mapping	2.1M	0.226	0.115	0.182	0.143	0.717	0.080	0.645	0.102
StyleSpace	6.0K	0.309	0.23	0.193	0.306	0.627	0.308	0.672	0.296
StyleSpaceSparse	**1.2K**	0.322	0.213	0.201	0.269	0.617	0.304	0.659	0.303

sual quality and low diversity in the generated images when considering Diversity metric. It indicates that for successful adaptation it is important to update the style vector from $\mathcal{S}$ rather than the intermediate latent vector from $\mathcal{W}$ space.

Analysis for few-shot domains. For this setting, as we discussed above, we consider two datasets, AFHQ Dogs and Cats [8], and the results are provided in Figure 4 and in Table 2. See more results in Appendix A.3.

We observe that results for Dogs and Cats are different compared to similar domains. In particular, we see that the Affine parameterization does not demonstrate the same quality as the Full parameterization. It can be seen from the degraded visual quality and visible gap in FID metric. However, it is still surprising that the generated images after adaptation have an adequate visual appearance considering that we do not fine-tune the synthesis network at all. SyntConv expectedly achieves results comparable with Full parameterization, and Mapping conversely shows poor quality on all datasets.

Table 2: FID scores for domain adaptation with different parameterizations. We observe a significant gap between Affine and Full parametrizations that, however, can be drastically reduced by introducing Affine+ parameterization.

		Domains	
Parameter Space	Size	Dog	Cat
Full	30.3M	20.3	7.1
SyntConv	23.6M	19.7	7.2
Affine	4.6M	70.1	27.6
Mapping	2.1M	208.2	226.1
Affine+	5.1M	**18.6**	**7.0**
AffineLight+	0.6M	20.6	8.9
StyleSpace	**6.0K**	75.8	22.0

4. Efficient and lightweight parameterizations of StyleGAN

StyleSpace and StyleSpaceSparse. Our findings from the previous section suggest that we can change the domain of generated images by modifying the style vector in the StyleSpace $\mathcal{S}$. To check this hypothesis, we will adapt StyleGAN2 by directly optimizing the direction in $\mathcal{S}$, i.e. during fine-tuning of StyleGAN we will optimize only Δs^D:

$$\mathcal{L}_D\left(\{G_\theta(s(z_i)+\Delta s)\}_{i=1}^K\right) \to \min_{\Delta s}, \tag{3}$$

where $\Delta s = (\Delta s_1, \ldots, \Delta s_N) \in \mathcal{S}$ is the optimized direction in the $\mathcal{S}$ for adapting the generator G_θ to the domain D. We call such directions Δs as *StyleDomain* directions.

Further, we explore that we can zero out most of the coordinates of StyleDomain directions without quality degradation. We use standard prunning technique when we leave 20% of the largest absolute values in the StyleDomain vector and set the rest to zero. We call such parameterization as *StyleSpaceSparse*. We examine different prunning rates and its performance in Appendix A.4.

We apply these parameterizations to one-shot and few-shot domains and obtain the following results.

For one-shot adaptation, we provide results in Figure 3 and in Table 1 (see more results in Appendix A.4). We observe that optimizing the StyleDomain direction Δs^D

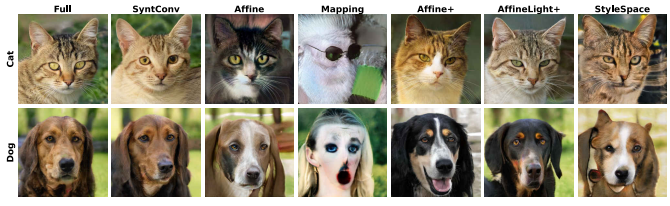


Figure 4: Domain adaptation for dissimilar domains. Affine+ parameterization produces results on par with the Full one.

achieves the same results both visually and quantitatively as the Full parameterization. It is new and important observation that StyleSpace allows not only image editing within domain but also generating samples from out-of domain of realistic human faces.

For few-shot domains, we observe that StyleSpace is not sufficient, which is expressed by the same significant quality degradation as for Affine parameterization. Further, we introduce a new parameterization that is efficient for more distant domains.

Affine+ and AffineLight+. We aim to improve Affine parameterization for successfully adapting to dissimilar domains such as Dogs and Cats. We propose to extend it by adding one block from synthesis layer with specified spatial resolution. Such block has two convolutional layers with weights $\theta_1, \theta_2 \in \mathbb{R}^{512\times512\times3\times3}$. Instead of fine-tuning all these weights we introduce more compact parameterization as offsets $\Delta\theta_1, \Delta\theta_2$ to these weights that are same across spatial dimensions, i.e. $\Delta\theta_1, \Delta\theta_2 \in \mathbb{R}^{512\times512\times1\times1}$ (we observe that such reduction in size does affect the performance). In addition to $\Delta\theta_1, \Delta\theta_2$ we introduce the offsets to the weights $\theta^{tRGB} \in \mathbb{R}^{3\times512\times3\times3}$ of the tRGB convolutional layer of the same block with similar parameterization $\Delta\theta^{tRGB} \in \mathbb{R}^{3\times512\times1\times1}$. Further, we omit such detail about tRGB part in the sake of brevity. So, for this parameterization the optimization procedure has the following form:

$$\mathcal{L}_D\left(\{G_{\theta,\Delta\theta_1,\Delta\theta_2}(s(z_i))\}_{i=1}^K\right) \rightarrow \min_{\Delta\theta_1,\Delta\theta_2,f^A}, \quad (4)$$

where $G_{\Delta\theta_1,\Delta\theta_2}$ is the generator with weight offsets $\Delta\theta_1, \Delta\theta_2$ for the one block from the synthesis network.

We call such parameter space as *Affine+*. We examine all blocks of the synthesis network to choose for this parameterization. We end up with the block with 64×64 resolution as it shows the best performance (see results of this analysis in Appendix A.4).

While Affine+ has already had 6 times less parameters than Full parameterization, we further reduce its size by applying low-rank decomposition to the weights of affine layers. We denote this parameterization as *AffineLight+*. It gives us by two orders less parameters than Full parameterization with insignificant degradation in quality. We will show in Section 5 that in low data regimes, this parameterization achieves comparable performance as Affine+ and outperforms other baselines. More details about applied low-rank decomposition can be found in Appendix A.4.

We apply these two parameterizations to few-shot domains and obtain the following results.

We provide results in Figure 4 and in Table 2 (see more results in Appendix A.4). We see that Affine+ parameterization allows us removing the performance gap from the Full parameterization both qualitatively and quantitatively. While the number of parameters for the additional block in Affine+ accounts for only 2 % of the synthesis network size. It shows that the style vector allows adapting the generator even to more distant domains if we just add a small part of the synthesis network.

We observe that AffineLight+ still shows adequate performance, except it has 100 times less parameters than the Full parameterization. We should notice that this parameterization is more suitable for low data regimes than those that we consider in Section 5.

Properties of the StyleDomain directions. We investigate StyleDomain directions that adapt the generator to similar domains and explore surprising properties. The first one is mixability, i.e. we discover that StyleDomain directions can be combined with each other. In particular, we can consider several directions that correspond to different similar domains and take their sum. The resulting direction will adapt the generator to the semantically mixed domain. We provide different examples of such combinations in Figure 6

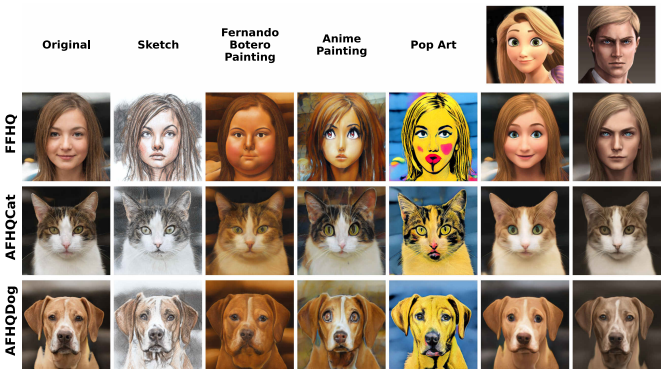


Figure 5: StyleSpace directions transfer from text-based and image-based domain adaptation to other fine-tuned models. We can successfully transfer style while preserving image content.



Figure 6: Example of mixing StyleDomain directions. We can combine different directions in order to perform adaptation into a semantically mixed domain.

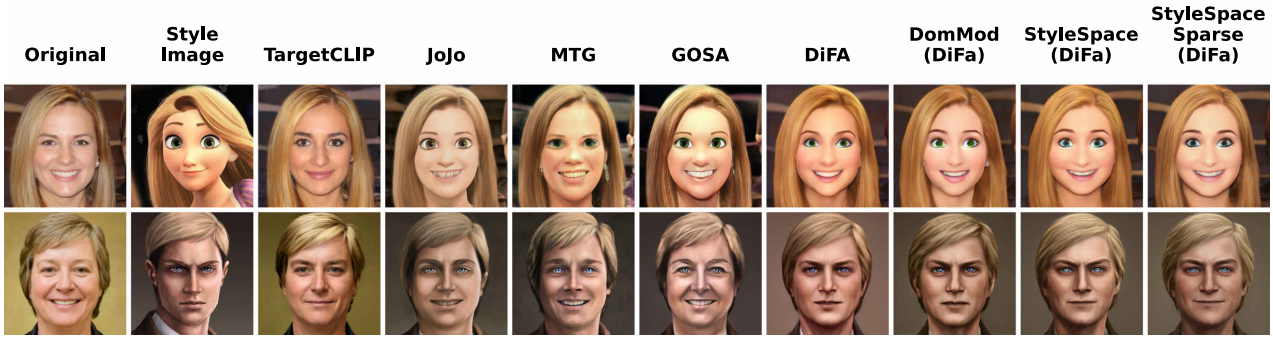


Figure 7: Comparison with baselines for one-shot image-based domain adaptation. StyleSpace and StyleSpaceSparse parameterizations achieve comparable quality as other methods while having much less trainable parameters.

(see more results in Appendix A.5).

The second property is transferability, i.e. we can transfer StyleDomain directions between different StyleGAN2 models. In particular, let us consider the base generator G_θ pretrained on realistic faces and the generator G_θ^{Dog} fine-tuned to domain of dogs in the Full parameterization. We verify that we can apply StyleDomain directions to G_θ^{Dog} which were optimized for G_θ. Specifically, we take StyleDomain directions that were optimized for G_θ to adapt it to different text-based and one-shot image-based domains (e.g. Pixar, Disney, etc.). Next, we apply these directions to generators that were fine-tuned from G_θ to other domains (e.g. Dogs, Cats). We provide results of this experiment in Figure 5 (see more results in Appendix A.5).

Finally, we check that StyleDomain directions can be successfully combined with latent controls for image editing. For a more detailed exploration of this property, see in Appendix A.5.

5. Experiments

One-shot domain adaptation. For image-based one-shot adaptation, we consider the main baselines: TargetCLIP [7], JoJoGAN [9], MTG [55], GOSA [49], DiFa [48], DomMod [4]. We apply our parameterizations StyleSpace and StyleSpaceSparse to DiFa model because this configuration achieves the highest performance (experiments with other base models can be found in Appendix A.6). Similarly, we apply DomMod parameterization to the DiFa model. We use StyleGAN2 as the GAN architecture and FFHQ as the source domain. For fair comparison, we strictly follow baseline default configurations in the choice of source-target adaptation setups and hyper-parameters. As target domain we use a wide range of different style images from baseline works. The full list of target images can be found in Appendix A.6.

We provide both quantitative and qualitative results in Table 3 and Figure 7, correspondingly. More results and

Table 3: Quality and Diversity metrics [4] for one-shot image-based domain adaptations with different methods. Memory denotes the memory needed for keeping adapted generators for all 12 domains for each method. StyleSpace and StyleSpaceSparse parameterizations achieve results comparable to other baselines while having much less trainable parameters.

			Titan Erwin		**Disney**		**Across 12 domains**	
Method	Size	Memory	Quality	Diversity	Quality	Diversity	Quality	Diversity
JoJoGAN [9]	30M	1.80GB	0.572	0.292	0.591	0.260	0.590 ± 0.048	0.257 ± 0.025
MTG [55]	30M	1.80GB	0.607	0.269	0.509	0.234	0.586 ± 0.054	0.263 ± 0.028
GOSA [49]	30M	1.80GB	0.547	0.283	0.617	0.216	0.584 ± 0.034	0.252 ± 0.030
DiFa [48]	30M	1.80GB	0.719	0.226	0.699	0.263	0.734 ± 0.047	0.215 ± 0.038
TargetCLIP [7]	9.0K	420KB	0.474	0.306	0.502	0.333	0.491 ± 0.043	0.322 ± 0.015
DomMod (DiFa) [4]	6.0K	280KB	0.705	0.250	0.625	0.294	0.679 ± 0.049	0.253 ± 0.037
StyleSpace (DiFa)	6.0K	280KB	0.672	0.296	0.627	0.308	0.644 ± 0.041	0.298 ± 0.025
StyleSpaceSparse (DiFa)	**1.2K**	**56.4KB**	0.659	0.303	0.617	0.304	0.638 ± 0.046	0.299 ± 0.026

samples can be found in Appendix A.6. We observe that the DiFa model achieves the best Quality metric but has low Diversity compared to other methods. Our parameterizations applied to the DiFa model balance its performance in terms of these metrics. So, StyleSpace (DiFa) and StyleSpaceSparse (DiFa) achieves uniformly better results than JoJoGAN, MTG, GOSA baselines. DomMod parameterization also show good performance, but it is comparable with our StyleSpaceSparse parameterization that has 5 times less parameters. We also report the overall memory needed for keeping adapted generators for 12 domains. We observe that StyleSpaceSparse requires significantly less space than other models. It will be especially important if we scale the number of target domains to thousands and more. We note that TargetCLIP that also has small number of trainable parameters show very poor results visually and in terms of Quality metric. It has a high Diversity only because its generated images are very close to the original ones. To provide more comprehensive comparison we additionally conduct user studies and present results in Appendix A.6.4.

Few-shot domain adaptation. As main baselines for this task, we consider the vanilla StyleGAN-ADA [18] (we will denote it as ADA), CDC [28] and recent SOTA method AdAM [51]. We compare our parameterizations Affine+ and AffineLight+ applied to ADA with these baselines on two datasets: Dogs and Cats [8]. We study the efficiency of these methods with different numbers of available target samples. For fair comparison, we rigorously follow the training setups of all baselines. We note that we use the same number of training iterations and the same batch size for all methods. We observe that in the AdAM paper [51] the number of training iterations is set to a small value (less than 10K) and it causes underfitting of the vanilla ADA method. Therefore, we increase this number for all methods to 50K and use the same batch size of 4. Then, for each method, we report the best FID value that it obtains during training. See more details about training in Appendix A.7.

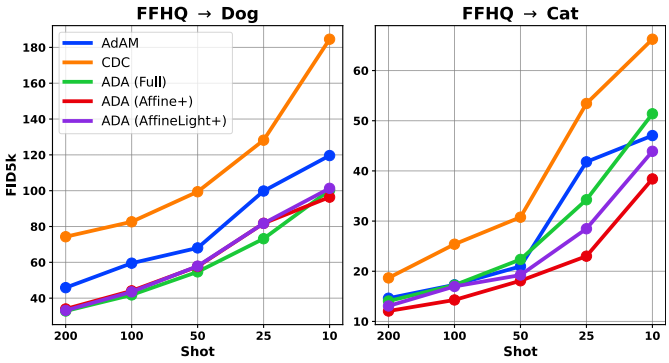


Figure 8: Few-shot training results for different number of shots. Proposed ADA (Affine+) and ADA (AffineLight+) show uniformly better performance than baselines.

Table 4: Results for few-shot training with 10-shots. Proposed ADA (Affine+) and ADA (AffineLight+) achieve better performance.

		Domains (10-shots)	
Method	Size	Cat	Dog
CDC [28]	30M	66.24	184.56
AdAM [51]	19M	47.05	119.61
ADA (Full) [18]	30M	51.38	100.25
ADA (Affine+)	5.1M	**38.40**	**96.38**
ADA (AffineLight+)	**0.6M**	43.91	101.31

We report results for few-shot training for different numbers of shots in Figure 8 and separately for 10 shots in Table 4. More results and samples can be found in Appendix A.7. Firstly, we observe that the AdAM method shows results not better than the vanilla ADA (Full) when trained for sufficient number of iterations. Secondly, we see that ADA (Affine+) and ADA (AffineLight+) based on our parameterizations achieve better results uniformly for all numbers of shots. It shows that these parameterizations is especially suitable for low data regimes.

Cross-domain image translation. We consider two setups of standard image-to-image problem. In the first one, we translate images from the source domain to the target domain unconditionally. In the second setup, we perform a reference-based translation, where the resulting image combines the pose and the shape of the source image with the style from a reference image. Details of these experiments can be found in Appendix A.9.

Cross-domain image morphing. Cross-domain morphing is a smooth transition between two images from different domains. This task is known as challenging [3, 10] and it is successfully tackled in the work [44] using aligned StyleGAN2 models. The idea is to interpolate between aligned generator weights to obtain a smooth transition between domains. We propose more complex image morphing by utilizing the transferability of StyleDomain directions. For example, we can apply a direction that stands for Sketch or Pixar style to Dogs domain to obtain a smooth transition between sketchy and pixar-like dog (see Figure 1). See many examples of such complex cross-domain morphing in Appendix A.10.

6. Conclusion

In this paper, we provide an extensive analysis of the process of StyleGAN domain adaptation. We reveal the sufficient components of the generator for successful adaptation depending on the similarity between the source and target domains. We discover the ability of StyleSpace to change

the domain of generated images by StyleDomain directions. We also propose new efficient parameterizations Affine+ and AffineLight+ for few-shot adaptation that outperform existing baselines. Further, we explore and leverage the properties of StyleDomain directions. We believe that our investigation can attract more attention to the exploration of new and interesting properties of StyleSpace.

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